

Midterm examination

20 questions 90 minutes 20 points

Name: _____

 Student ID (three digits): _____ Form: **M7KD**

Instructions

- Mark exactly one answer A–G per question on this response page. Each correct answer earns one point; incorrect and blank answers earn zero. No negative marking.
- Work independently. Notes, calculators, and lecture slides are allowed. Approved accommodations remain valid.
- Fill all three ID digits and each answer bubble clearly. Keep the QR code and corner marks clear.
- Question pages follow this response page. You may use the booklet margins for working. All numerical examples are hypothetical.

Notation and supplied normal values $N(m, v)$ has mean m and *variance* v . OLS includes an intercept unless stated otherwise. SSR is the residual sum of squares; ESS is explained sum of squares. Φ is the standard normal CDF and $\Phi(-z) = 1 - \Phi(z)$.

$$\Phi(1) = 0.84134, \quad \Phi(2) = 0.97725, \quad \Phi(3) = 0.99865, \quad \Phi(4) = 0.9999683.$$

Student ID: fill one digit in each column

Digit 1	Digit 2	Digit 3
☐ 0	☐ 0	☐ 0
☐ 1	☐ 1	☐ 1
☐ 2	☐ 2	☐ 2
☐ 3	☐ 3	☐ 3
☐ 4	☐ 4	☐ 4
☐ 5	☐ 5	☐ 5
☐ 6	☐ 6	☐ 6
☐ 7	☐ 7	☐ 7
☐ 8	☐ 8	☐ 8
☐ 9	☐ 9	☐ 9

Use your assigned three-digit numeric student ID. Fill the matching circle in each column, including any leading zero. An incomplete or unreadable ID can prevent grading.

Answers: fill one circle per row

	A	B	C	D	E	F	G
1	☐	☐	☐	☐	☐	☐	☐
2	☐	☐	☐	☐	☐	☐	☐
3	☐	☐	☐	☐	☐	☐	☐
4	☐	☐	☐	☐	☐	☐	☐
5	☐	☐	☐	☐	☐	☐	☐
6	☐	☐	☐	☐	☐	☐	☐
7	☐	☐	☐	☐	☐	☐	☐
8	☐	☐	☐	☐	☐	☐	☐
9	☐	☐	☐	☐	☐	☐	☐
10	☐	☐	☐	☐	☐	☐	☐

	A	B	C	D	E	F	G
11	☐	☐	☐	☐	☐	☐	☐
12	☐	☐	☐	☐	☐	☐	☐
13	☐	☐	☐	☐	☐	☐	☐
14	☐	☐	☐	☐	☐	☐	☐
15	☐	☐	☐	☐	☐	☐	☐
16	☐	☐	☐	☐	☐	☐	☐
17	☐	☐	☐	☐	☐	☐	☐
18	☐	☐	☐	☐	☐	☐	☐
19	☐	☐	☐	☐	☐	☐	☐
20	☐	☐	☐	☐	☐	☐	☐

Medium questions 1–10

1. The entries below are joint probabilities.

	$Y = 0$	$Y = 2$
$X = 0$	0.30	0.10
$X = 1$	0.20	0.40

Which pair gives $(E(Y | X = 1), E(Y))$?

- A. $(\frac{4}{3}, 2)$ B. $(\frac{4}{5}, \frac{6}{5})$ C. $(1, \frac{4}{3})$ D. $(\frac{4}{5}, \frac{4}{5})$
 E. $(\frac{4}{3}, \frac{7}{6})$ F. $(\frac{4}{5}, 1)$ G. $(\frac{4}{3}, 1)$

2. Y takes 0, 2, 4 with probabilities 1/4, 1/2, 1/4. Define $W = 7 - 2Y$. Which pair gives $(E(W), \text{Var}(W))$?

- A. (11, 8) B. (4, 8) C. (3, 8) D. (3, 15)
 E. (3, -8) F. (3, 4) G. (3, 24)

3. $\text{SD}(X) = 2$, $\text{SD}(Y) = 4$, and $\text{Corr}(X, Y) = -1/2$. Let $W = 5 + 2X$ and $V = 9 - 3Y$. Which pair gives $(\text{Cov}(W, V), \text{Corr}(W, V))$?

- A. (24, 3) B. $(69, \frac{1}{2})$ C. $(-24, -\frac{1}{2})$ D. $(-4, -\frac{1}{2})$
 E. $(24, -\frac{1}{2})$ F. $(24, \frac{1}{2})$ G. $(4, \frac{1}{2})$

4. Three i.i.d. observations have mean μ and variance 4. Compare $A = (Y_1 + Y_2 + Y_3)/3$ and $B = Y_1/2 + Y_2/4 + Y_3/4$. Which pair gives $(\text{Var}(B), \text{Var}(B)/\text{Var}(A))$?

- A. $(2, \frac{3}{2})$ B. $(\frac{4}{3}, 1)$ C. $(\frac{3}{2}, \frac{9}{8})$ D. $(\frac{3}{2}, \frac{1}{8})$
 E. (4, 3) F. $(1, \frac{3}{4})$ G. $(\frac{3}{2}, \frac{8}{9})$

5. Let Y_i be i.i.d. with mean μ and finite positive variance. Define $A_n = Y_1$, $B_n = \bar{Y}_n + 2/n$, and $C_n = \bar{Y}_n + 2$. Which statement correctly classifies all three rules for estimating μ ?

- A. All three are consistent because the nominal sample size grows.
 B. All three are inconsistent because finite-sample bias is nonzero.
 C. A is unbiased and inconsistent; B is biased and consistent; C is biased and inconsistent.
 D. A is unbiased and consistent; B is biased and inconsistent; C is biased and inconsistent.
 E. A is biased and inconsistent; B is unbiased and consistent; C is biased and inconsistent.
 F. A is unbiased and inconsistent; B and C are biased and consistent.
 G. A and B are unbiased and consistent; C is biased and inconsistent.

6. Four i.i.d. observations are 3, 5, 7, 9; the population shape is unspecified. Use $s_Y^2 = \sum_i (Y_i - \bar{Y})^2 / (n-1)$ and $SE = s_Y / \sqrt{n}$. Which pair gives (s_Y^2, SE^2) ?

- A. $(5, \frac{5}{4})$ B. $(\frac{20}{3}, \frac{5}{12})$ C. $(\frac{20}{3}, \frac{10}{3})$ D. $(\frac{20}{3}, \frac{20}{3})$
 E. $(\frac{20}{3}, \frac{5}{3})$ F. $(\frac{5}{3}, \frac{20}{3})$ G. (20, 5)

7. An i.i.d. sample from a population with finite positive variance has $n = 144$, $\bar{y} = 22$ and sample SD $s_Y = 12$. Test $H_0 : \mu = 20$ against $H_1 : \mu \neq 20$ using the large-sample normal approximation. Which option gives the statistic, two-sided p-value, and decision at a prespecified 5% level? Use $\Phi(2) = 0.97725$.

- A. $t = 2$, $p \approx 0.0455$; do not reject. B. $t = 24$, $p \approx 0.0455$; reject.
 C. $t = 2$, $p \approx 0.02275$; reject. D. $t = 2$, $p \approx 0.0455$; reject.
 E. $t = -2$, $p \approx 0.0455$; reject. F. $t = 1/6$, $p \approx 0.0455$; reject.
 G. $t = 2$, $p \approx 0.9545$; do not reject.

8. OLS with an intercept uses $\bar{X} = 3$, $\bar{Y} = 9$, $S_{XX} = \sum_i (X_i - \bar{X})^2 = 10$, and $S_{XY} = \sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = 20$. Which pair gives $(\hat{\beta}_0, \hat{\beta}_1)$?

- A. (3, 20) B. (2, 3) C. (3, -2) D. (3, 2)
 E. (3, $\frac{1}{2}$) F. (15, 2) G. (9, 2)

9. A simple OLS regression with an intercept uses $n = 18$ observations and reports $SSR = 64$ and $TSS = 256$. Here SSR is the residual sum of squares. Which pair gives (R^2, SER) ?

- A. ($\frac{3}{4}$, 2) B. ($\frac{3}{4}$, 16) C. ($\frac{3}{4}$, $2\sqrt{8/9}$) D. ($\frac{1}{3}$, 2)
 E. ($\frac{3}{4}$, $2\sqrt{16/17}$) F. ($\frac{3}{4}$, 4) G. ($\frac{1}{4}$, 2)

10. Maintain the causal model $Y = \alpha + 3X + 2Z + \varepsilon$, with $E(\varepsilon | X, Z) = 0$, $\text{Var}(X) = 4$ and $\text{Cov}(X, Z) = -2$. Pairs are sampled i.i.d. with finite fourth moments. Let $\hat{b}_s$ be the OLS slope on X alone, with an intercept. Which pair gives $(b_s, \text{plim } \hat{b}_s)$?

- A. (3, 2) B. (1, 1) C. (2, 3) D. (3, 3)
 E. (4, 4) F. (2, 2) G. (-1, -1)

Hard questions 11–20

11. The table lists joint probabilities, and $E(Y) = 0.90$.

	$Y = 0$	$Y = 3$	Total
$X = 0$	0.30	0.10	0.40
$X = 1$	a	b	0.60

Which triple gives $(a, b, \text{Pr}(X = 1 | Y = 3))$?

- A. ($\frac{2}{5}$, $\frac{1}{5}$, $\frac{3}{5}$) B. ($\frac{2}{5}$, $\frac{1}{5}$, $\frac{3}{2}$) C. ($\frac{2}{5}$, $\frac{1}{5}$, $\frac{3}{10}$) D. ($\frac{2}{5}$, $\frac{1}{5}$, $\frac{1}{5}$)
 E. ($\frac{1}{2}$, $\frac{1}{10}$, $\frac{2}{3}$) F. ($\frac{2}{5}$, $\frac{1}{5}$, $\frac{1}{3}$) G. ($\frac{2}{5}$, $\frac{1}{5}$, $\frac{2}{3}$)

12. For finite supports write $p_x = \text{Pr}(X = x) > 0$, $p_y = \text{Pr}(Y = y)$, and $p_{xy} = \text{Pr}(X = x, Y = y)$. A proof uses

$$E[E(Y | X)] = \sum_x \sum_y y \text{Pr}(Y = y | X = x) \boxed{1} = \sum_y y \sum_x \boxed{2} = E(Y).$$

Which choice supplies both blanks and states the required dependence assumption? Each replacement must be valid *term by term for every* (x, y) in an arbitrary joint distribution.

- A. $(1, \text{Pr}(Y = y | X = x))$; independence is unnecessary.
 B. (p_y, p_{xy}) ; independence is unnecessary.
 C. (p_x, p_y) ; independence is unnecessary.
 D. (p_x, p_{xy}) ; independence is unnecessary.
 E. $(p_x, p_x p_y)$; independence is unnecessary.
 F. (p_x, p_{xy}) ; independence is required.
 G. $(1/p_x, p_{xy})$; independence is unnecessary.

13. X takes $-3, -2, 2, 3$ with equal probabilities. Define $u = X^2 - \frac{13}{2}$. Which choice correctly gives $E(Xu)$, $E(X^2u)$, and whether $E(u | X) = 0$ holds?

- A. $(0, -\frac{25}{4})$; zero conditional mean fails. B. $(0, \frac{25}{4})$; zero conditional mean fails.
 C. $(0, \frac{25}{2})$; zero conditional mean fails. D. $(0, 0)$; zero conditional mean holds.
 E. $(\frac{25}{4}, 0)$; zero conditional mean fails. F. $(0, \frac{13}{2})$; zero conditional mean fails.
 G. $(0, \frac{25}{4})$; zero conditional mean holds.

14. Let Y_i be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. For $n > 4$, let $\bar{Y}_{-1}$ average observations $2, \dots, n$, and define

$$T_n = \frac{1}{2}Y_1 + \frac{1}{2}\bar{Y}_{-1}, \quad S_n = \frac{2}{\sqrt{n}}Y_1 + \left(1 - \frac{2}{\sqrt{n}}\right)\bar{Y}_{-1}.$$

Which pair gives $(\lim \text{Var}(T_n)/\sigma^2, \lim \text{Var}(S_n)/(\sigma^2/n))$ as $n \rightarrow \infty$?

- A. $(\frac{1}{4}, 5)$ B. $(\frac{1}{2}, 5)$ C. $(\frac{1}{4}, 0)$ D. $(\frac{1}{4}, 4)$
 E. $(0, 1)$ F. $(0, 5)$ G. $(\frac{1}{4}, 1)$

15. Independent Y_1, Y_2 are Bernoulli(0.20). Set $w_1 = 1$ if $Y_1 \geq Y_2$ and $w_1 = 0$ otherwise; $w_2 = 1 - w_1$. Write $T = w_1Y_1 + w_2Y_2$. A student claims

$$E(T) \stackrel{(1)}{=} \sum_{i=1}^2 E(w_i Y_i) \stackrel{(2)}{=} \sum_{i=1}^2 E(w_i) E(Y_i) \stackrel{(3)}{=} 0.20.$$

Which choice gives the actual $E(T)$ and the first invalid step?

- A. 0.04; step (2), because weights depend on outcomes.
 B. 0.20; no step is invalid.
 C. 0.36; step (2), because Y_1, Y_2 are dependent.
 D. 0.80; step (2), because weights depend on outcomes.
 E. 0.36; step (2), because weights depend on outcomes.
 F. 0.36; step (1), because expectation is not linear.
 G. 0.36; step (3), because weights sum to two.

16. Let Y_i be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Define $B_n = \bar{Y}_n - 1\sigma/\sqrt{n}$. Which pair gives its probability limit and the limiting distribution of $\sqrt{n}(B_n - \mu)/\sigma$? The second normal parameter is variance.

- A. $(\mu, \text{diverges to } -\infty)$ B. $(\mu, N(1, 1))$ C. $(\mu, \text{constant zero})$ D. $(\mu, N(-1, 2))$
 E. $(\mu, N(0, 1))$ F. $(\mu - 1\sigma, N(-1, 1))$ G. $(\mu, N(-1, 1))$

17. An i.i.d. sample from a normal population has $n = 100$, $\bar{y} = 12$ and known population SD $\sigma = 10$. Test $H_0 : \mu = 10$ against $H_1 : \mu \neq 10$. Each record is copied four times, creating 400 rows from the same 100 independent draws. Which choice gives the *correct* z-statistic, two-sided p-value, and decision at 1% after copying? Use $\Phi(2) = 0.97725$ and $\Phi(4) = 0.9999683$.

- A. $z = 4$, $p \approx 0.0000634$; reject. B. $z = 2$, $p \approx 0.0455$; do not reject.
 C. $z = 4$, $p \approx 0.0455$; do not reject. D. $z = 2$, $p \approx 0.0455$; reject.
 E. $z = 2$, $p \approx 0.9545$; do not reject. F. $z = 2$, $p \approx 0.02275$; do not reject.
 G. $z = 1$, $p \approx 0.0455$; do not reject.

18. OLS with an intercept gives $\hat{Y} = 3 + 2X$ for

$$X = (0, 1, 2, 3), \quad Y = (5, a_*, b_*, 11).$$

Which triple gives (a_*, b_*, R^2) ?

- A. $(5, 3, \frac{5}{9})$ B. $(3, 5, \frac{5}{9})$ C. $(5, 7, 1)$ D. $(1, 7, \frac{5}{9})$
 E. $(3, 5, \frac{5}{4})$ F. $(3, 5, \frac{4}{9})$ G. $(3, 5, 1)$

19. For an arbitrary simple OLS sample *with an intercept*, assume $S_{XX} > 0$ and $TSS > 0$. Squaring $Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + \hat{u}_i$ gives

$$TSS = ESS + SSR + 2C, \quad C = \sum_i (\hat{Y}_i - \bar{Y}) \hat{u}_i.$$

Which expression for C, together with its justification for being zero, is valid for every such sample?

- A.** $\hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$; both sample normal equations.
- B.** $\sum_i \hat{u}_i^2$; the residuals have zero sample mean.
- C.** $\sum_i (Y_i - \bar{Y})^2$; an intercept makes total variation zero.
- D.** $\hat{\beta}_1[\sum_i X_i u_i - \bar{X} \sum_i u_i]$; the sample normal equations for population errors.
- E.** $\hat{\beta}_1^2 S_{XX}$; the slope normal equation.
- F.** $\hat{\beta}_1 \sum_i X_i \hat{u}_i$; zero population conditional mean must be assumed.
- G.** $\hat{\beta}_1 \sum_i (X_i - \bar{X})^2$; the intercept normal equation.

20. Maintain $Y = \alpha + \beta X + Z + \varepsilon$, with $E(\varepsilon | X, Z) = 0$, $SD(X) = SD(Z) = 2$, and $\text{Corr}(X, Z) = 1/2$. The population short-regression slope on X alone is $b_s = 1$. Under i.i.d. sampling and the lecture's finite-moment conditions,

$$\sqrt{n}(\hat{b}_s - b_s) \xrightarrow{d} N(0, 25).$$

Let $P_n = \Pr(|\hat{b}_s - \beta| \leq 1/4)$. Which triple gives $(\beta, P_{400}, \lim_{n \rightarrow \infty} P_n)$? Use the stated large-sample approximation, $\Phi(1) = 0.84134$ and $\Phi(3) = 0.99865$.

- | | | | |
|--|--|--|--|
| A. $(\frac{1}{2}, \approx 0.6827, 1)$ | B. $(\frac{1}{2}, \approx 0.9987, 0)$ | C. $(\frac{3}{2}, \approx 0.1573, 0)$ | D. $(\frac{1}{2}, \approx 0.1573, 1)$ |
| E. $(\frac{1}{2}, \approx 0.1573, 0)$ | F. $(1, \approx 0.6827, 1)$ | G. $(\frac{1}{2}, \approx 0.3173, 0)$ | |

Midterm examination

20 questions 90 minutes 20 points

Name: _____

Student ID (three digits): _____ Form: M3RX



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$$\Phi(1) = 0.84134, \quad \Phi(2) = 0.97725, \quad \Phi(3) = 0.99865, \quad \Phi(4) = 0.9999683.$$

Student ID: fill one digit in each column

Digit 1	Digit 2	Digit 3
①	①	①
②	②	②
③	③	③
④	④	④
⑤	⑤	⑤
⑥	⑥	⑥
⑦	⑦	⑦
⑧	⑧	⑧
⑨	⑨	⑨

Use your assigned three-digit numeric student ID. Fill the matching circle in each column, including any leading zero. An incomplete or unreadable ID can prevent grading.

Answers: fill one circle per row

	A	B	C	D	E	F	G
1	○	○	○	○	○	○	○
2	○	○	○	○	○	○	○
3	○	○	○	○	○	○	○
4	○	○	○	○	○	○	○
5	○	○	○	○	○	○	○
6	○	○	○	○	○	○	○
7	○	○	○	○	○	○	○
8	○	○	○	○	○	○	○
9	○	○	○	○	○	○	○
10	○	○	○	○	○	○	○

	A	B	C	D	E	F	G
11	○	○	○	○	○	○	○
12	○	○	○	○	○	○	○
13	○	○	○	○	○	○	○
14	○	○	○	○	○	○	○
15	○	○	○	○	○	○	○
16	○	○	○	○	○	○	○
17	○	○	○	○	○	○	○
18	○	○	○	○	○	○	○
19	○	○	○	○	○	○	○
20	○	○	○	○	○	○	○

Medium questions 1–10

1. The entries below are joint probabilities.

	$Y = 0$	$Y = 3$
$X = 0$	0.30	0.10
$X = 1$	0.20	0.40

Which pair gives $(E(Y | X = 1), E(Y))$?

- A. $(\frac{6}{5}, \frac{3}{2})$ B. $(2, 3)$ C. $(\frac{3}{2}, 2)$ D. $(\frac{6}{5}, \frac{6}{5})$
 E. $(2, \frac{3}{2})$ F. $(2, \frac{7}{4})$ G. $(\frac{6}{5}, \frac{9}{5})$

2. Y takes 0, 2, 4 with probabilities $1/4, 1/2, 1/4$. Define $W = 10 - 3Y$. Which pair gives $(E(W), \text{Var}(W))$?

- A. (4, 6) B. (16, 18) C. (4, 54) D. (4, 28)
 E. (6, 18) F. (4, -18) G. (4, 18)

3. $\text{SD}(X) = 2$, $\text{SD}(Y) = 5$, and $\text{Corr}(X, Y) = -1/2$. Let $W = 5 + 2X$ and $V = 9 - 3Y$. Which pair gives $(\text{Cov}(W, V), \text{Corr}(W, V))$?

- A. $(-30, -\frac{1}{2})$ B. $(30, \frac{1}{2})$ C. $(75, \frac{1}{2})$ D. $(30, -\frac{1}{2})$
 E. (30, 3) F. $(5, \frac{1}{2})$ G. $(-5, -\frac{1}{2})$

4. Three i.i.d. observations have mean μ and variance 9. Compare $A = (Y_1 + Y_2 + Y_3)/3$ and $B = Y_1/2 + Y_2/4 + Y_3/4$. Which pair gives $(\text{Var}(B), \text{Var}(B)/\text{Var}(A))$?

- A. (9, 3) B. (3, 1) C. $(\frac{9}{4}, \frac{3}{4})$ D. $(\frac{27}{8}, \frac{1}{8})$
 E. $(\frac{9}{2}, \frac{3}{2})$ F. $(\frac{27}{8}, \frac{9}{8})$ G. $(\frac{27}{8}, \frac{8}{9})$

5. Let Y_i be i.i.d. with mean μ and finite positive variance. Define $A_n = Y_1$, $B_n = \bar{Y}_n + 3/n$, and $C_n = \bar{Y}_n + 3$. Which statement correctly classifies all three rules for estimating μ ?

- A. A is unbiased and consistent; B is biased and inconsistent; C is biased and inconsistent.
 B. All three are inconsistent because finite-sample bias is nonzero.
 C. A is biased and inconsistent; B is unbiased and consistent; C is biased and inconsistent.
 D. A is unbiased and inconsistent; B and C are biased and consistent.
 E. A is unbiased and inconsistent; B is biased and consistent; C is biased and inconsistent.
 F. All three are consistent because the nominal sample size grows.
 G. A and B are unbiased and consistent; C is biased and inconsistent.

6. Four i.i.d. observations are 4, 6, 8, 10; the population shape is unspecified. Use $s_Y^2 = \sum_i (Y_i - \bar{Y})^2 / (n - 1)$ and $SE = s_Y / \sqrt{n}$. Which pair gives (s_Y^2, SE^2) ?

- A. $(\frac{20}{3}, \frac{5}{3})$ B. $(\frac{5}{3}, \frac{20}{3})$ C. $(\frac{20}{3}, \frac{5}{12})$ D. $(\frac{20}{3}, \frac{10}{3})$
 E. (20, 5) F. $(\frac{20}{3}, \frac{20}{3})$ G. $(5, \frac{5}{4})$

7. An i.i.d. sample from a population with finite positive variance has $n = 144$, $\bar{y} = 23$ and sample SD $s_Y = 12$. Test $H_0 : \mu = 21$ against $H_1 : \mu \neq 21$ using the large-sample normal approximation. Which option gives the statistic, two-sided p-value, and decision at a prespecified 5% level? Use $\Phi(2) = 0.97725$.

- A. $t = 2$, $p \approx 0.0455$; reject. B. $t = 2$, $p \approx 0.02275$; reject.
 C. $t = 1/6$, $p \approx 0.0455$; reject. D. $t = -2$, $p \approx 0.0455$; reject.
 E. $t = 2$, $p \approx 0.9545$; do not reject. F. $t = 2$, $p \approx 0.0455$; do not reject.
 G. $t = 24$, $p \approx 0.0455$; reject.

8. OLS with an intercept uses $\bar{X} = 3$, $\bar{Y} = 13$, $S_{XX} = \sum_i (X_i - \bar{X})^2 = 10$, and $S_{XY} = \sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = 30$. Which pair gives $(\hat{\beta}_0, \hat{\beta}_1)$?

- A. (4, -3) B. (4, 3) C. (13, 3) D. (3, 4)
 E. (22, 3) F. (4, $\frac{1}{3}$) G. (4, 30)

9. A simple OLS regression with an intercept uses $n = 18$ observations and reports $SSR = 144$ and $TSS = 576$. Here SSR is the residual sum of squares. Which pair gives (R^2, SER) ?

- A. ($\frac{1}{3}$, 3) B. ($\frac{3}{4}$, $3\sqrt{8/9}$) C. ($\frac{3}{4}$, 9) D. ($\frac{3}{4}$, 3)
 E. ($\frac{1}{4}$, 3) F. ($\frac{3}{4}$, $3\sqrt{16/17}$) G. ($\frac{3}{4}$, 24)

10. Maintain the causal model $Y = \alpha + 4X + 2Z + \varepsilon$, with $E(\varepsilon | X, Z) = 0$, $\text{Var}(X) = 4$ and $\text{Cov}(X, Z) = -2$. Pairs are sampled i.i.d. with finite fourth moments. Let $\hat{b}_s$ be the OLS slope on X alone, with an intercept. Which pair gives $(b_s, \text{plim } \hat{b}_s)$?

- A. (3, 4) B. (4, 3) C. (0, 0) D. (4, 4)
 E. (2, 2) F. (5, 5) G. (3, 3)

Hard questions 11–20

11. The table lists joint probabilities, and $E(Y) = 1.40$.

	$Y = 0$	$Y = 4$	Total
$X = 0$	0.30	0.10	0.40
$X = 1$	a	b	0.60

Which triple gives $(a, b, \text{Pr}(X = 1 | Y = 4))$?

- A. ($\frac{7}{20}$, $\frac{1}{4}$, $\frac{1}{4}$) B. ($\frac{7}{20}$, $\frac{1}{4}$, $\frac{5}{12}$) C. ($\frac{7}{20}$, $\frac{1}{4}$, $\frac{5}{7}$) D. ($\frac{7}{20}$, $\frac{1}{4}$, $\frac{7}{5}$)
 E. ($\frac{7}{20}$, $\frac{1}{4}$, $\frac{3}{5}$) F. ($\frac{9}{20}$, $\frac{3}{20}$, $\frac{5}{7}$) G. ($\frac{7}{20}$, $\frac{1}{4}$, $\frac{7}{20}$)

12. For finite supports write $p_x = \text{Pr}(X = x) > 0$, $p_y = \text{Pr}(Y = y)$, and $p_{xy} = \text{Pr}(X = x, Y = y)$. A proof uses

$$E[E(Y | X)] = \sum_x \sum_y y \text{Pr}(Y = y | X = x) \boxed{1} = \sum_y y \sum_x \boxed{2} = E(Y).$$

Which choice supplies both blanks and states the required dependence assumption? Each replacement must be valid *term by term for every* (x, y) in an arbitrary joint distribution.

- A. (p_y, p_{xy}) ; independence is unnecessary.
 B. $(1/p_x, p_{xy})$; independence is unnecessary.
 C. (p_x, p_y) ; independence is unnecessary.
 D. $(1, \text{Pr}(Y = y | X = x))$; independence is unnecessary.
 E. (p_x, p_{xy}) ; independence is unnecessary.
 F. (p_x, p_{xy}) ; independence is required.
 G. $(p_x, p_x p_y)$; independence is unnecessary.

13. X takes $-4, -3, 3, 4$ with equal probabilities. Define $u = X^2 - \frac{25}{2}$. Which choice correctly gives $E(Xu)$, $E(X^2u)$, and whether $E(u | X) = 0$ holds?

- A. $(0, -\frac{49}{4})$; zero conditional mean fails. B. $(0, \frac{49}{2})$; zero conditional mean fails.
 C. $(0, \frac{25}{2})$; zero conditional mean fails. D. $(0, \frac{49}{4})$; zero conditional mean fails.
 E. $(\frac{49}{4}, 0)$; zero conditional mean fails. F. $(0, 0)$; zero conditional mean holds.
 G. $(0, \frac{49}{4})$; zero conditional mean holds.

14. Let Y_i be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. For $n > 4$, let $\bar{Y}_{-1}$ average observations $2, \dots, n$, and define

$$T_n = \frac{1}{3}Y_1 + \frac{2}{3}\bar{Y}_{-1}, \quad S_n = \frac{3}{\sqrt{n}}Y_1 + \left(1 - \frac{3}{\sqrt{n}}\right)\bar{Y}_{-1}.$$

Which pair gives $(\lim \text{Var}(T_n)/\sigma^2, \lim \text{Var}(S_n)/(\sigma^2/n))$ as $n \rightarrow \infty$?

- A. (0, 1) B. $(\frac{1}{9}, 9)$ C. $(\frac{1}{9}, 1)$ D. $(\frac{1}{9}, 0)$
 E. $(\frac{1}{3}, 10)$ F. $(\frac{1}{9}, 10)$ G. (0, 10)

15. Independent Y_1, Y_2 are Bernoulli(0.30). Set $w_1 = 1$ if $Y_1 \geq Y_2$ and $w_1 = 0$ otherwise; $w_2 = 1 - w_1$. Write $T = w_1Y_1 + w_2Y_2$. A student claims

$$E(T) \stackrel{(1)}{=} \sum_{i=1}^2 E(w_i Y_i) \stackrel{(2)}{=} \sum_{i=1}^2 E(w_i) E(Y_i) \stackrel{(3)}{=} 0.30.$$

Which choice gives the actual $E(T)$ and the first invalid step?

- A. 0.30; no step is invalid.
 B. 0.51; step (1), because expectation is not linear.
 C. 0.51; step (2), because weights depend on outcomes.
 D. 0.51; step (3), because weights sum to two.
 E. 0.09; step (2), because weights depend on outcomes.
 F. 0.70; step (2), because weights depend on outcomes.
 G. 0.51; step (2), because Y_1, Y_2 are dependent.

16. Let Y_i be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Define $B_n = \bar{Y}_n - 2\sigma/\sqrt{n}$. Which pair gives its probability limit and the limiting distribution of $\sqrt{n}(B_n - \mu)/\sigma$? The second normal parameter is variance.

- A. $(\mu, N(-2, 5))$ B. $(\mu, \text{diverges to } -\infty)$ C. $(\mu, N(-2, 1))$ D. $(\mu, N(0, 1))$
 E. $(\mu, \text{constant zero})$ F. $(\mu, N(2, 1))$ G. $(\mu - 2\sigma, N(-2, 1))$

17. An i.i.d. sample from a normal population has $n = 144$, $\bar{y} = 13$ and known population SD $\sigma = 12$. Test $H_0 : \mu = 11$ against $H_1 : \mu \neq 11$. Each record is copied four times, creating 576 rows from the same 144 independent draws. Which choice gives the *correct* z-statistic, two-sided p-value, and decision at 1% after copying? Use $\Phi(2) = 0.97725$ and $\Phi(4) = 0.9999683$.

- A. $z = 2$, $p \approx 0.9545$; do not reject. B. $z = 2$, $p \approx 0.0455$; reject.
 C. $z = 2$, $p \approx 0.02275$; do not reject. D. $z = 2$, $p \approx 0.0455$; do not reject.
 E. $z = 4$, $p \approx 0.0000634$; reject. F. $z = 1$, $p \approx 0.0455$; do not reject.
 G. $z = 4$, $p \approx 0.0455$; do not reject.

18. OLS with an intercept gives $\hat{Y} = 4 + 3X$ for

$$X = (0, 1, 2, 3), \quad Y = (7, a_*, b_*, 16).$$

Which triple gives (a_*, b_*, R^2) ?

- A. $(4, 7, \frac{5}{9})$ B. $(4, 7, 1)$ C. $(4, 7, \frac{4}{9})$ D. $(1, 10, \frac{5}{9})$
 E. $(7, 4, \frac{5}{9})$ F. $(4, 7, \frac{5}{4})$ G. $(7, 10, 1)$

19. For an arbitrary simple OLS sample *with an intercept*, assume $S_{XX} > 0$ and $TSS > 0$. Squaring $Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + \hat{u}_i$ gives

$$TSS = ESS + SSR + 2C, \quad C = \sum_i (\hat{Y}_i - \bar{Y}) \hat{u}_i.$$

Which expression for C, together with its justification for being zero, is valid for every such sample?

- A.** $\hat{\beta}_1^2 S_{XX}$; the slope normal equation.
- B.** $\sum_i \hat{u}_i^2$; the residuals have zero sample mean.
- C.** $\hat{\beta}_1 [\sum_i X_i u_i - \bar{X} \sum_i u_i]$; the sample normal equations for population errors.
- D.** $\hat{\beta}_1 \sum_i (X_i - \bar{X})^2$; the intercept normal equation.
- E.** $\sum_i (Y_i - \bar{Y})^2$; an intercept makes total variation zero.
- F.** $\hat{\beta}_1 [\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$; both sample normal equations.
- G.** $\hat{\beta}_1 \sum_i X_i \hat{u}_i$; zero population conditional mean must be assumed.

20. Maintain $Y = \alpha + \beta X + Z + \varepsilon$, with $E(\varepsilon | X, Z) = 0$, $SD(X) = SD(Z) = 2$, and $\text{Corr}(X, Z) = 1/2$. The population short-regression slope on X alone is $b_s = 2$. Under i.i.d. sampling and the lecture's finite-moment conditions,

$$\sqrt{n}(\hat{b}_s - b_s) \xrightarrow{d} N(0, 25).$$

Let $P_n = \Pr(|\hat{b}_s - \beta| \leq 1/4)$. Which triple gives $(\beta, P_{400}, \lim_{n \rightarrow \infty} P_n)$? Use the stated large-sample approximation, $\Phi(1) = 0.84134$ and $\Phi(3) = 0.99865$.

- | | | | |
|--|--|--|--|
| A. $(2, \approx 0.6827, 1)$ | B. $(\frac{5}{2}, \approx 0.1573, 0)$ | C. $(\frac{3}{2}, \approx 0.1573, 1)$ | D. $(\frac{3}{2}, \approx 0.9987, 0)$ |
| E. $(\frac{3}{2}, \approx 0.6827, 1)$ | F. $(\frac{3}{2}, \approx 0.3173, 0)$ | G. $(\frac{3}{2}, \approx 0.1573, 0)$ | |

Midterm examination

20 questions 90 minutes 20 points

Name: _____

Student ID (three digits): _____ Form: M8VN



Instructions

- Mark exactly one answer A–G per question on this response page. Each correct answer earns one point; incorrect and blank answers earn zero. No negative marking.
- Work independently. Notes, calculators, and lecture slides are allowed. Approved accommodations remain valid.
- Fill all three ID digits and each answer bubble clearly. Keep the QR code and corner marks clear.
- Question pages follow this response page. You may use the booklet margins for working. All numerical examples are hypothetical.

Notation and supplied normal values $N(m, v)$ has mean m and variance v . OLS includes an intercept unless stated otherwise. SSR is the residual sum of squares; ESS is explained sum of squares. Φ is the standard normal CDF and $\Phi(-z) = 1 - \Phi(z)$.

$$\Phi(1) = 0.84134, \quad \Phi(2) = 0.97725, \quad \Phi(3) = 0.99865, \quad \Phi(4) = 0.9999683.$$

Student ID: fill one digit in each column

Digit 1	Digit 2	Digit 3
①	①	①
②	②	②
③	③	③
④	④	④
⑤	⑤	⑤
⑥	⑥	⑥
⑦	⑦	⑦
⑧	⑧	⑧
⑨	⑨	⑨

Use your assigned three-digit numeric student ID. Fill the matching circle in each column, including any leading zero. An incomplete or unreadable ID can prevent grading.

Answers: fill one circle per row

	A	B	C	D	E	F	G
1	○	○	○	○	○	○	○
2	○	○	○	○	○	○	○
3	○	○	○	○	○	○	○
4	○	○	○	○	○	○	○
5	○	○	○	○	○	○	○
6	○	○	○	○	○	○	○
7	○	○	○	○	○	○	○
8	○	○	○	○	○	○	○
9	○	○	○	○	○	○	○
10	○	○	○	○	○	○	○

	A	B	C	D	E	F	G
11	○	○	○	○	○	○	○
12	○	○	○	○	○	○	○
13	○	○	○	○	○	○	○
14	○	○	○	○	○	○	○
15	○	○	○	○	○	○	○
16	○	○	○	○	○	○	○
17	○	○	○	○	○	○	○
18	○	○	○	○	○	○	○
19	○	○	○	○	○	○	○
20	○	○	○	○	○	○	○

Medium questions 1–10

1. The entries below are joint probabilities.

	$Y = 0$	$Y = 4$
$X = 0$	0.30	0.10
$X = 1$	0.20	0.40

Which pair gives $(E(Y | X = 1), E(Y))$?

- A. $(\frac{8}{5}, \frac{8}{5})$ B. $(\frac{8}{5}, \frac{12}{5})$ C. $(\frac{8}{3}, 2)$ D. $(\frac{8}{5}, 2)$
 E. $(2, \frac{8}{3})$ F. $(\frac{8}{3}, 4)$ G. $(\frac{8}{3}, \frac{7}{3})$

2. Y takes 0, 2, 4 with probabilities $1/4, 1/2, 1/4$. Define $W = 13 - 4Y$. Which pair gives $(E(W), \text{Var}(W))$?

- A. (5, 8) B. (5, 32) C. (5, 96) D. (8, 32)
 E. (5, -32) F. (21, 32) G. (5, 45)

3. $\text{SD}(X) = 2$, $\text{SD}(Y) = 6$, and $\text{Corr}(X, Y) = -1/2$. Let $W = 5 + 2X$ and $V = 9 - 3Y$. Which pair gives $(\text{Cov}(W, V), \text{Corr}(W, V))$?

- A. $(36, \frac{1}{2})$ B. $(-6, -\frac{1}{2})$ C. $(81, \frac{1}{2})$ D. $(36, -\frac{1}{2})$
 E. $(6, \frac{1}{2})$ F. $(36, 3)$ G. $(-36, -\frac{1}{2})$

4. Three i.i.d. observations have mean μ and variance 16. Compare $A = (Y_1 + Y_2 + Y_3)/3$ and $B = Y_1/2 + Y_2/4 + Y_3/4$. Which pair gives $(\text{Var}(B), \text{Var}(B)/\text{Var}(A))$?

- A. $(6, \frac{1}{8})$ B. $(4, \frac{3}{4})$ C. $(\frac{16}{3}, 1)$ D. $(6, \frac{8}{9})$
 E. $(16, 3)$ F. $(8, \frac{3}{2})$ G. $(6, \frac{9}{8})$

5. Let Y_i be i.i.d. with mean μ and finite positive variance. Define $A_n = Y_1$, $B_n = \bar{Y}_n + 4/n$, and $C_n = \bar{Y}_n + 4$. Which statement correctly classifies all three rules for estimating μ ?

- A. A is unbiased and consistent; B is biased and inconsistent; C is biased and inconsistent.
 B. All three are inconsistent because finite-sample bias is nonzero.
 C. A is unbiased and inconsistent; B and C are biased and consistent.
 D. A is biased and inconsistent; B is unbiased and consistent; C is biased and inconsistent.
 E. All three are consistent because the nominal sample size grows.
 F. A is unbiased and inconsistent; B is biased and consistent; C is biased and inconsistent.
 G. A and B are unbiased and consistent; C is biased and inconsistent.

6. Four i.i.d. observations are 5, 7, 9, 11; the population shape is unspecified. Use $s_Y^2 = \sum_i (Y_i - \bar{Y})^2 / (n - 1)$ and $SE = s_Y / \sqrt{n}$. Which pair gives (s_Y^2, SE^2) ?

- A. $(5, \frac{5}{4})$ B. $(\frac{20}{3}, \frac{5}{3})$ C. (20, 5) D. $(\frac{5}{3}, \frac{20}{3})$
 E. $(\frac{20}{3}, \frac{5}{12})$ F. $(\frac{20}{3}, \frac{20}{3})$ G. $(\frac{20}{3}, \frac{10}{3})$

7. An i.i.d. sample from a population with finite positive variance has $n = 144$, $\bar{y} = 24$ and sample SD $s_Y = 12$. Test $H_0 : \mu = 22$ against $H_1 : \mu \neq 22$ using the large-sample normal approximation. Which option gives the statistic, two-sided p-value, and decision at a prespecified 5% level? Use $\Phi(2) = 0.97725$.

- A. $t = 1/6$, $p \approx 0.0455$; reject. B. $t = 24$, $p \approx 0.0455$; reject.
 C. $t = 2$, $p \approx 0.0455$; do not reject. D. $t = 2$, $p \approx 0.9545$; do not reject.
 E. $t = 2$, $p \approx 0.0455$; reject. F. $t = -2$, $p \approx 0.0455$; reject.
 G. $t = 2$, $p \approx 0.02275$; reject.

8. OLS with an intercept uses $\bar{X} = 3$, $\bar{Y} = 17$, $S_{XX} = \sum_i (X_i - \bar{X})^2 = 10$, and $S_{XY} = \sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = 40$. Which pair gives $(\hat{\beta}_0, \hat{\beta}_1)$?

- A. (5, -4) B. (29, 4) C. (5, 40) D. (17, 4)
 E. (4, 5) F. (5, 4) G. (5, $\frac{1}{4}$)

9. A simple OLS regression with an intercept uses $n = 18$ observations and reports $SSR = 256$ and $TSS = 1024$. Here SSR is the residual sum of squares. Which pair gives (R^2, SER) ?

- A. ($\frac{3}{4}$, $4\sqrt{16/17}$) B. ($\frac{1}{3}$, 4) C. ($\frac{3}{4}$, 16) D. ($\frac{1}{4}$, 4)
 E. ($\frac{3}{4}$, 32) F. ($\frac{3}{4}$, $4\sqrt{8/9}$) G. ($\frac{3}{4}$, 4)

10. Maintain the causal model $Y = \alpha + 5X + 2Z + \varepsilon$, with $E(\varepsilon | X, Z) = 0$, $\text{Var}(X) = 4$ and $\text{Cov}(X, Z) = -2$. Pairs are sampled i.i.d. with finite fourth moments. Let $\hat{b}_s$ be the OLS slope on X alone, with an intercept. Which pair gives $(b_s, \text{plim } \hat{b}_s)$?

- A. (1, 1) B. (5, 5) C. (5, 4) D. (4, 4)
 E. (4, 5) F. (6, 6) G. (3, 3)

Hard questions 11–20

11. The table lists joint probabilities, and $E(Y) = 2.00$.

	Y = 0	Y = 5	Total
X = 0	0.30	0.10	0.40
X = 1	a	b	0.60

Which triple gives $(a, b, \text{Pr}(X = 1 | Y = 5))$?

- A. ($\frac{3}{10}$, $\frac{3}{10}$, $\frac{2}{5}$) B. ($\frac{3}{10}$, $\frac{3}{10}$, $\frac{3}{5}$) C. ($\frac{3}{10}$, $\frac{3}{10}$, $\frac{4}{3}$) D. ($\frac{3}{10}$, $\frac{3}{10}$, $\frac{3}{10}$)
 E. ($\frac{3}{10}$, $\frac{3}{10}$, $\frac{3}{4}$) F. ($\frac{2}{5}$, $\frac{1}{5}$, $\frac{3}{4}$) G. ($\frac{3}{10}$, $\frac{3}{10}$, $\frac{1}{2}$)

12. For finite supports write $p_x = \text{Pr}(X = x) > 0$, $p_y = \text{Pr}(Y = y)$, and $p_{xy} = \text{Pr}(X = x, Y = y)$. A proof uses

$$E[E(Y | X)] = \sum_x \sum_y y \text{Pr}(Y = y | X = x) \boxed{1} = \sum_y y \sum_x \boxed{2} = E(Y).$$

Which choice supplies both blanks and states the required dependence assumption? Each replacement must be valid *term by term for every* (x, y) in an arbitrary joint distribution.

- A. (p_y, p_{xy}) ; independence is unnecessary.
 B. $(p_x, p_x p_y)$; independence is unnecessary.
 C. (p_x, p_{xy}) ; independence is unnecessary.
 D. (p_x, p_{xy}) ; independence is required.
 E. (p_x, p_y) ; independence is unnecessary.
 F. $(1, \text{Pr}(Y = y | X = x))$; independence is unnecessary.
 G. $(1/p_x, p_{xy})$; independence is unnecessary.

13. X takes $-5, -4, 4, 5$ with equal probabilities. Define $u = X^2 - \frac{41}{2}$. Which choice correctly gives $E(Xu)$, $E(X^2u)$, and whether $E(u | X) = 0$ holds?

- A. $(0, -\frac{81}{4})$; zero conditional mean fails. B. $(0, \frac{41}{2})$; zero conditional mean fails.
 C. $(0, \frac{81}{4})$; zero conditional mean fails. D. $(\frac{81}{4}, 0)$; zero conditional mean fails.
 E. $(0, \frac{81}{4})$; zero conditional mean holds. F. $(0, 0)$; zero conditional mean holds.
 G. $(0, \frac{81}{2})$; zero conditional mean fails.

14. Let Y_i be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. For $n > 4$, let $\bar{Y}_{-1}$ average observations $2, \dots, n$, and define

$$T_n = \frac{2}{3}Y_1 + \frac{1}{3}\bar{Y}_{-1}, \quad S_n = \frac{2}{\sqrt{n}}Y_1 + \left(1 - \frac{2}{\sqrt{n}}\right)\bar{Y}_{-1}.$$

Which pair gives $(\lim \text{Var}(T_n)/\sigma^2, \lim \text{Var}(S_n)/(\sigma^2/n))$ as $n \rightarrow \infty$?

- A. $(\frac{2}{3}, 5)$ B. $(\frac{4}{9}, 1)$ C. $(0, 5)$ D. $(\frac{4}{9}, 5)$
 E. $(\frac{4}{9}, 4)$ F. $(\frac{4}{9}, 0)$ G. $(0, 1)$

15. Independent Y_1, Y_2 are Bernoulli(0.40). Set $w_1 = 1$ if $Y_1 \geq Y_2$ and $w_1 = 0$ otherwise; $w_2 = 1 - w_1$. Write $T = w_1Y_1 + w_2Y_2$. A student claims

$$E(T) \stackrel{(1)}{=} \sum_{i=1}^2 E(w_i Y_i) \stackrel{(2)}{=} \sum_{i=1}^2 E(w_i) E(Y_i) \stackrel{(3)}{=} 0.40.$$

Which choice gives the actual $E(T)$ and the first invalid step?

- A. 0.60; step (2), because weights depend on outcomes.
 B. 0.64; step (1), because expectation is not linear.
 C. 0.16; step (2), because weights depend on outcomes.
 D. 0.64; step (3), because weights sum to two.
 E. 0.40; no step is invalid.
 F. 0.64; step (2), because weights depend on outcomes.
 G. 0.64; step (2), because Y_1, Y_2 are dependent.

16. Let Y_i be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Define $B_n = \bar{Y}_n - 3\sigma/\sqrt{n}$. Which pair gives its probability limit and the limiting distribution of $\sqrt{n}(B_n - \mu)/\sigma$? The second normal parameter is variance.

- A. $(\mu, N(-3, 1))$ B. $(\mu, N(-3, 10))$ C. $(\mu, N(0, 1))$ D. $(\mu, \text{constant zero})$
 E. $(\mu, \text{diverges to } -\infty)$ F. $(\mu - 3\sigma, N(-3, 1))$ G. $(\mu, N(3, 1))$

17. An i.i.d. sample from a normal population has $n = 196$, $\bar{y} = 14$ and known population SD $\sigma = 14$. Test $H_0 : \mu = 12$ against $H_1 : \mu \neq 12$. Each record is copied four times, creating 784 rows from the same 196 independent draws. Which choice gives the *correct* z-statistic, two-sided p-value, and decision at 1% after copying? Use $\Phi(2) = 0.97725$ and $\Phi(4) = 0.9999683$.

- A. $z = 4$, $p \approx 0.0000634$; reject. B. $z = 2$, $p \approx 0.0455$; reject.
 C. $z = 1$, $p \approx 0.0455$; do not reject. D. $z = 2$, $p \approx 0.9545$; do not reject.
 E. $z = 4$, $p \approx 0.0455$; do not reject. F. $z = 2$, $p \approx 0.02275$; do not reject.
 G. $z = 2$, $p \approx 0.0455$; do not reject.

18. OLS with an intercept gives $\hat{Y} = 5 + 4X$ for

$$X = (0, 1, 2, 3), \quad Y = (9, a_*, b_*, 21).$$

Which triple gives (a_*, b_*, R^2) ?

- A. $(5, 9, \frac{4}{9})$ B. $(5, 9, 1)$ C. $(9, 13, 1)$ D. $(1, 13, \frac{5}{9})$
 E. $(5, 9, \frac{5}{9})$ F. $(9, 5, \frac{5}{9})$ G. $(5, 9, \frac{5}{4})$

19. For an arbitrary simple OLS sample *with an intercept*, assume $S_{XX} > 0$ and $TSS > 0$. Squaring $Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + \hat{u}_i$ gives

$$TSS = ESS + SSR + 2C, \quad C = \sum_i (\hat{Y}_i - \bar{Y}) \hat{u}_i.$$

Which expression for C, together with its justification for being zero, is valid for every such sample?

- A. $\sum_i (Y_i - \bar{Y})^2$; an intercept makes total variation zero.
- B. $\hat{\beta}_1 [\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$; both sample normal equations.
- C. $\hat{\beta}_1 \sum_i (X_i - \bar{X})^2$; the intercept normal equation.
- D. $\hat{\beta}_1 \sum_i X_i \hat{u}_i$; zero population conditional mean must be assumed.
- E. $\sum_i \hat{u}_i^2$; the residuals have zero sample mean.
- F. $\hat{\beta}_1^2 S_{XX}$; the slope normal equation.
- G. $\hat{\beta}_1 [\sum_i X_i u_i - \bar{X} \sum_i u_i]$; the sample normal equations for population errors.

20. Maintain $Y = \alpha + \beta X + Z + \varepsilon$, with $E(\varepsilon | X, Z) = 0$, $SD(X) = SD(Z) = 2$, and $\text{Corr}(X, Z) = 1/2$. The population short-regression slope on X alone is $b_s = 3$. Under i.i.d. sampling and the lecture's finite-moment conditions,

$$\sqrt{n}(\hat{b}_s - b_s) \xrightarrow{d} N(0, 25).$$

Let $P_n = \Pr(|\hat{b}_s - \beta| \leq 1/4)$. Which triple gives $(\beta, P_{400}, \lim_{n \rightarrow \infty} P_n)$? Use the stated large-sample approximation, $\Phi(1) = 0.84134$ and $\Phi(3) = 0.99865$.

- | | | | |
|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| A. $(\frac{5}{2}, \approx 0.1573, 0)$ | B. $(\frac{5}{2}, \approx 0.3173, 0)$ | C. $(\frac{5}{2}, \approx 0.9987, 0)$ | D. $(\frac{5}{2}, \approx 0.6827, 1)$ |
| E. $(3, \approx 0.6827, 1)$ | F. $(\frac{5}{2}, \approx 0.1573, 1)$ | G. $(\frac{7}{2}, \approx 0.1573, 0)$ | |

Midterm examination

20 questions 90 minutes 20 points

Name: _____

 Student ID (three digits): _____ Form: **M2HF**

Instructions

- Mark exactly one answer A–G per question on this response page. Each correct answer earns one point; incorrect and blank answers earn zero. No negative marking.
- Work independently. Notes, calculators, and lecture slides are allowed. Approved accommodations remain valid.
- Fill all three ID digits and each answer bubble clearly. Keep the QR code and corner marks clear.
- Question pages follow this response page. You may use the booklet margins for working. All numerical examples are hypothetical.

Notation and supplied normal values $N(m, v)$ has mean m and *variance* v . OLS includes an intercept unless stated otherwise. SSR is the residual sum of squares; ESS is explained sum of squares. Φ is the standard normal CDF and $\Phi(-z) = 1 - \Phi(z)$.

$$\Phi(1) = 0.84134, \quad \Phi(2) = 0.97725, \quad \Phi(3) = 0.99865, \quad \Phi(4) = 0.9999683.$$

Student ID: fill one digit in each column

Digit 1	Digit 2	Digit 3
☐ 0	☐ 0	☐ 0
☐ 1	☐ 1	☐ 1
☐ 2	☐ 2	☐ 2
☐ 3	☐ 3	☐ 3
☐ 4	☐ 4	☐ 4
☐ 5	☐ 5	☐ 5
☐ 6	☐ 6	☐ 6
☐ 7	☐ 7	☐ 7
☐ 8	☐ 8	☐ 8
☐ 9	☐ 9	☐ 9

Use your assigned three-digit numeric student ID. Fill the matching circle in each column, including any leading zero. An incomplete or unreadable ID can prevent grading.

Answers: fill one circle per row

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Medium questions 1–10

1. The entries below are joint probabilities.

	$Y = 0$	$Y = 5$
$X = 0$	0.30	0.10
$X = 1$	0.20	0.40

Which pair gives $(E(Y | X = 1), E(Y))$?

- A. $(2, \frac{5}{2})$ B. $(2, 3)$ C. $(\frac{5}{2}, \frac{10}{3})$ D. $(\frac{10}{3}, 5)$
 E. $(2, 2)$ F. $(\frac{10}{3}, \frac{5}{2})$ G. $(\frac{10}{3}, \frac{35}{12})$

2. Y takes 0, 2, 4 with probabilities $1/4, 1/2, 1/4$. Define $W = 16 - 5Y$. Which pair gives $(E(W), \text{Var}(W))$?

- A. (26, 50) B. (6, 150) C. (6, 10) D. (6, 50)
 E. (6, 66) F. (6, -50) G. (10, 50)

3. $\text{SD}(X) = 2$, $\text{SD}(Y) = 7$, and $\text{Corr}(X, Y) = -1/2$. Let $W = 5 + 2X$ and $V = 9 - 3Y$. Which pair gives $(\text{Cov}(W, V), \text{Corr}(W, V))$?

- A. (42, 3) B. $(-7, -\frac{1}{2})$ C. $(42, -\frac{1}{2})$ D. $(7, \frac{1}{2})$
 E. $(42, \frac{1}{2})$ F. $(-42, -\frac{1}{2})$ G. $(87, \frac{1}{2})$

4. Three i.i.d. observations have mean μ and variance 25. Compare $A = (Y_1 + Y_2 + Y_3)/3$ and $B = Y_1/2 + Y_2/4 + Y_3/4$. Which pair gives $(\text{Var}(B), \text{Var}(B)/\text{Var}(A))$?

- A. $(\frac{75}{8}, \frac{9}{8})$ B. (25, 3) C. $(\frac{75}{8}, \frac{1}{8})$ D. $(\frac{25}{3}, 1)$
 E. $(\frac{25}{2}, \frac{3}{2})$ F. $(\frac{25}{4}, \frac{3}{4})$ G. $(\frac{75}{8}, \frac{8}{9})$

5. Let Y_i be i.i.d. with mean μ and finite positive variance. Define $A_n = Y_1$, $B_n = \bar{Y}_n + 5/n$, and $C_n = \bar{Y}_n + 5$. Which statement correctly classifies all three rules for estimating μ ?

- A. A is biased and inconsistent; B is unbiased and consistent; C is biased and inconsistent.
 B. All three are consistent because the nominal sample size grows.
 C. A is unbiased and consistent; B is biased and inconsistent; C is biased and inconsistent.
 D. All three are inconsistent because finite-sample bias is nonzero.
 E. A and B are unbiased and consistent; C is biased and inconsistent.
 F. A is unbiased and inconsistent; B and C are biased and consistent.
 G. A is unbiased and inconsistent; B is biased and consistent; C is biased and inconsistent.

6. Four i.i.d. observations are 6, 8, 10, 12; the population shape is unspecified. Use $s_Y^2 = \sum_i (Y_i - \bar{Y})^2 / (n - 1)$ and $SE = s_Y / \sqrt{n}$. Which pair gives (s_Y^2, SE^2) ?

- A. $(\frac{20}{3}, \frac{20}{3})$ B. (20, 5) C. $(\frac{5}{3}, \frac{20}{3})$ D. $(\frac{20}{3}, \frac{5}{3})$
 E. $(\frac{20}{3}, \frac{10}{3})$ F. $(5, \frac{5}{4})$ G. $(\frac{20}{3}, \frac{5}{12})$

7. An i.i.d. sample from a population with finite positive variance has $n = 144$, $\bar{y} = 25$ and sample SD $s_Y = 12$. Test $H_0 : \mu = 23$ against $H_1 : \mu \neq 23$ using the large-sample normal approximation. Which option gives the statistic, two-sided p-value, and decision at a prespecified 5% level? Use $\Phi(2) = 0.97725$.

- A. $t = 1/6$, $p \approx 0.0455$; reject. B. $t = 2$, $p \approx 0.0455$; reject.
 C. $t = 24$, $p \approx 0.0455$; reject. D. $t = 2$, $p \approx 0.0455$; do not reject.
 E. $t = 2$, $p \approx 0.9545$; do not reject. F. $t = 2$, $p \approx 0.02275$; reject.
 G. $t = -2$, $p \approx 0.0455$; reject.

8. OLS with an intercept uses $\bar{X} = 3$, $\bar{Y} = 21$, $S_{XX} = \sum_i (X_i - \bar{X})^2 = 10$, and $S_{XY} = \sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = 50$. Which pair gives $(\hat{\beta}_0, \hat{\beta}_1)$?

- A. (6, 5) B. $(6, \frac{1}{5})$ C. (5, 6) D. (36, 5)
 E. (21, 5) F. (6, -5) G. (6, 50)

9. A simple OLS regression with an intercept uses $n = 18$ observations and reports $SSR = 400$ and $TSS = 1600$. Here SSR is the residual sum of squares. Which pair gives (R^2, SER) ?

- A. $(\frac{1}{3}, 5)$ B. $(\frac{3}{4}, 5\sqrt{8/9})$ C. $(\frac{3}{4}, 5\sqrt{16/17})$ D. $(\frac{3}{4}, 25)$
 E. $(\frac{3}{4}, 5)$ F. $(\frac{1}{4}, 5)$ G. $(\frac{3}{4}, 40)$

10. Maintain the causal model $Y = \alpha + 6X + 2Z + \varepsilon$, with $E(\varepsilon | X, Z) = 0$, $\text{Var}(X) = 4$ and $\text{Cov}(X, Z) = -2$. Pairs are sampled i.i.d. with finite fourth moments. Let $\hat{b}_s$ be the OLS slope on X alone, with an intercept. Which pair gives $(b_s, \text{plim } \hat{b}_s)$?

- A. (5, 5) B. (6, 6) C. (2, 2) D. (6, 5)
 E. (5, 6) F. (4, 4) G. (7, 7)

Hard questions 11–20

11. The table lists joint probabilities, and $E(Y) = 2.70$.

	$Y = 0$	$Y = 6$	Total
$X = 0$	0.30	0.10	0.40
$X = 1$	a	b	0.60

Which triple gives $(a, b, \text{Pr}(X = 1 | Y = 6))$?

- A. $(\frac{7}{20}, \frac{1}{4}, \frac{7}{9})$ B. $(\frac{1}{4}, \frac{7}{20}, \frac{7}{9})$ C. $(\frac{1}{4}, \frac{7}{20}, \frac{9}{7})$ D. $(\frac{1}{4}, \frac{7}{20}, \frac{9}{20})$
 E. $(\frac{1}{4}, \frac{7}{20}, \frac{7}{12})$ F. $(\frac{1}{4}, \frac{7}{20}, \frac{3}{5})$ G. $(\frac{1}{4}, \frac{7}{20}, \frac{7}{20})$

12. For finite supports write $p_x = \text{Pr}(X = x) > 0$, $p_y = \text{Pr}(Y = y)$, and $p_{xy} = \text{Pr}(X = x, Y = y)$. A proof uses

$$E[E(Y | X)] = \sum_x \sum_y y \text{Pr}(Y = y | X = x) \boxed{1} = \sum_y y \sum_x \boxed{2} = E(Y).$$

Which choice supplies both blanks and states the required dependence assumption? Each replacement must be valid *term by term for every* (x, y) in an arbitrary joint distribution.

- A. $(1, \text{Pr}(Y = y | X = x))$; independence is unnecessary.
 B. $(p_x, p_x p_y)$; independence is unnecessary.
 C. (p_y, p_{xy}) ; independence is unnecessary.
 D. $(1/p_x, p_{xy})$; independence is unnecessary.
 E. (p_x, p_y) ; independence is unnecessary.
 F. (p_x, p_{xy}) ; independence is required.
 G. (p_x, p_{xy}) ; independence is unnecessary.

13. X takes $-6, -5, 5, 6$ with equal probabilities. Define $u = X^2 - \frac{61}{2}$. Which choice correctly gives $E(Xu)$, $E(X^2u)$, and whether $E(u | X) = 0$ holds?

- A. $(0, \frac{121}{4})$; zero conditional mean holds. B. $(0, \frac{121}{2})$; zero conditional mean fails.
 C. $(0, -\frac{121}{4})$; zero conditional mean fails. D. $(0, 0)$; zero conditional mean holds.
 E. $(0, \frac{121}{4})$; zero conditional mean fails. F. $(0, \frac{61}{2})$; zero conditional mean fails.
 G. $(\frac{121}{4}, 0)$; zero conditional mean fails.

14. Let Y_i be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. For $n > 4$, let $\bar{Y}_{-1}$ average observations $2, \dots, n$, and define

$$T_n = \frac{1}{4}Y_1 + \frac{3}{4}\bar{Y}_{-1}, \quad S_n = \frac{3}{\sqrt{n}}Y_1 + \left(1 - \frac{3}{\sqrt{n}}\right)\bar{Y}_{-1}.$$

Which pair gives $(\lim \text{Var}(T_n)/\sigma^2, \lim \text{Var}(S_n)/(\sigma^2/n))$ as $n \rightarrow \infty$?

- A. $(\frac{1}{16}, 9)$ B. $(0, 1)$ C. $(\frac{1}{16}, 0)$ D. $(\frac{1}{16}, 1)$
 E. $(\frac{1}{4}, 10)$ F. $(0, 10)$ G. $(\frac{1}{16}, 10)$

15. Independent Y_1, Y_2 are Bernoulli(0.60). Set $w_1 = 1$ if $Y_1 \geq Y_2$ and $w_1 = 0$ otherwise; $w_2 = 1 - w_1$. Write $T = w_1Y_1 + w_2Y_2$. A student claims

$$E(T) \stackrel{(1)}{=} \sum_{i=1}^2 E(w_i Y_i) \stackrel{(2)}{=} \sum_{i=1}^2 E(w_i) E(Y_i) \stackrel{(3)}{=} 0.60.$$

Which choice gives the actual $E(T)$ and the first invalid step?

- A. 0.40; step (2), because weights depend on outcomes.
 B. 0.84; step (2), because weights depend on outcomes.
 C. 0.60; no step is invalid.
 D. 0.84; step (2), because Y_1, Y_2 are dependent.
 E. 0.84; step (3), because weights sum to two.
 F. 0.84; step (1), because expectation is not linear.
 G. 0.36; step (2), because weights depend on outcomes.

16. Let Y_i be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Define $B_n = \bar{Y}_n - 4\sigma/\sqrt{n}$. Which pair gives its probability limit and the limiting distribution of $\sqrt{n}(B_n - \mu)/\sigma$? The second normal parameter is variance.

- A. $(\mu, N(4, 1))$ B. $(\mu, N(-4, 17))$ C. $(\mu - 4\sigma, N(-4, 1))$ D. $(\mu, N(0, 1))$
 E. $(\mu, \text{diverges to } -\infty)$ F. $(\mu, N(-4, 1))$ G. $(\mu, \text{constant zero})$

17. An i.i.d. sample from a normal population has $n = 256$, $\bar{y} = 15$ and known population SD $\sigma = 16$. Test $H_0 : \mu = 13$ against $H_1 : \mu \neq 13$. Each record is copied four times, creating 1024 rows from the same 256 independent draws. Which choice gives the *correct* z-statistic, two-sided p-value, and decision at 1% after copying? Use $\Phi(2) = 0.97725$ and $\Phi(4) = 0.9999683$.

- A. $z = 4, p \approx 0.0455$; do not reject. B. $z = 2, p \approx 0.9545$; do not reject.
 C. $z = 2, p \approx 0.0455$; do not reject. D. $z = 4, p \approx 0.0000634$; reject.
 E. $z = 1, p \approx 0.0455$; do not reject. F. $z = 2, p \approx 0.02275$; do not reject.
 G. $z = 2, p \approx 0.0455$; reject.

18. OLS with an intercept gives $\hat{Y} = 6 + 5X$ for

$$X = (0, 1, 2, 3), \quad Y = (11, a_*, b_*, 26).$$

Which triple gives (a_*, b_*, R^2) ?

- A. $(1, 16, \frac{5}{9})$ B. $(11, 6, \frac{5}{9})$ C. $(6, 11, \frac{5}{9})$ D. $(6, 11, 1)$
 E. $(6, 11, \frac{5}{4})$ F. $(11, 16, 1)$ G. $(6, 11, \frac{4}{9})$

19. For an arbitrary simple OLS sample *with an intercept*, assume $S_{XX} > 0$ and $TSS > 0$. Squaring $Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + \hat{u}_i$ gives

$$TSS = ESS + SSR + 2C, \quad C = \sum_i (\hat{Y}_i - \bar{Y}) \hat{u}_i.$$

Which expression for C, together with its justification for being zero, is valid for every such sample?

- A.** $\hat{\beta}_1 [\sum_i X_i u_i - \bar{X} \sum_i u_i]$; the sample normal equations for population errors.
- B.** $\hat{\beta}_1^2 S_{XX}$; the slope normal equation.
- C.** $\hat{\beta}_1 \sum_i X_i \hat{u}_i$; zero population conditional mean must be assumed.
- D.** $\hat{\beta}_1 [\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$; both sample normal equations.
- E.** $\sum_i (Y_i - \bar{Y})^2$; an intercept makes total variation zero.
- F.** $\sum_i \hat{u}_i^2$; the residuals have zero sample mean.
- G.** $\hat{\beta}_1 \sum_i (X_i - \bar{X})^2$; the intercept normal equation.

20. Maintain $Y = \alpha + \beta X + Z + \varepsilon$, with $E(\varepsilon | X, Z) = 0$, $SD(X) = SD(Z) = 2$, and $\text{Corr}(X, Z) = 1/2$. The population short-regression slope on X alone is $b_s = 4$. Under i.i.d. sampling and the lecture's finite-moment conditions,

$$\sqrt{n}(\hat{b}_s - b_s) \xrightarrow{d} N(0, 25).$$

Let $P_n = \Pr(|\hat{b}_s - \beta| \leq 1/4)$. Which triple gives $(\beta, P_{400}, \lim_{n \rightarrow \infty} P_n)$? Use the stated large-sample approximation, $\Phi(1) = 0.84134$ and $\Phi(3) = 0.99865$.

- | | | | |
|--|--|--|------------------------------------|
| A. $(\frac{7}{2}, \approx 0.3173, 0)$ | B. $(\frac{7}{2}, \approx 0.9987, 0)$ | C. $(\frac{7}{2}, \approx 0.6827, 1)$ | D. $(4, \approx 0.6827, 1)$ |
| E. $(\frac{9}{2}, \approx 0.1573, 0)$ | F. $(\frac{7}{2}, \approx 0.1573, 0)$ | G. $(\frac{7}{2}, \approx 0.1573, 1)$ | |

Midterm examination

20 questions 90 minutes 20 points

Name: _____

 Student ID (three digits): _____ Form: **M6WP**

Instructions

- Mark exactly one answer A–G per question on this response page. Each correct answer earns one point; incorrect and blank answers earn zero. No negative marking.
- Work independently. Notes, calculators, and lecture slides are allowed. Approved accommodations remain valid.
- Fill all three ID digits and each answer bubble clearly. Keep the QR code and corner marks clear.
- Question pages follow this response page. You may use the booklet margins for working. All numerical examples are hypothetical.

Notation and supplied normal values $N(m, v)$ has mean m and *variance* v . OLS includes an intercept unless stated otherwise. SSR is the residual sum of squares; ESS is explained sum of squares. Φ is the standard normal CDF and $\Phi(-z) = 1 - \Phi(z)$.

$$\Phi(1) = 0.84134, \quad \Phi(2) = 0.97725, \quad \Phi(3) = 0.99865, \quad \Phi(4) = 0.9999683.$$

Student ID: fill one digit in each column

Digit 1	Digit 2	Digit 3
☐ 0	☐ 0	☐ 0
☐ 1	☐ 1	☐ 1
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☐ 3	☐ 3	☐ 3
☐ 4	☐ 4	☐ 4
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☐ 9	☐ 9	☐ 9

Use your assigned three-digit numeric student ID. Fill the matching circle in each column, including any leading zero. An incomplete or unreadable ID can prevent grading.

Answers: fill one circle per row

	A	B	C	D	E	F	G
1	☐	☐	☐	☐	☐	☐	☐
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6	☐	☐	☐	☐	☐	☐	☐
7	☐	☐	☐	☐	☐	☐	☐
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	A	B	C	D	E	F	G
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18	☐	☐	☐	☐	☐	☐	☐
19	☐	☐	☐	☐	☐	☐	☐
20	☐	☐	☐	☐	☐	☐	☐

Medium questions 1–10

1. The entries below are joint probabilities.

	$Y = 0$	$Y = 6$
$X = 0$	0.30	0.10
$X = 1$	0.20	0.40

Which pair gives $(E(Y | X = 1), E(Y))$?

- A. $(4, \frac{7}{2})$ B. $(4, 3)$ C. $(4, 6)$ D. $(\frac{12}{5}, \frac{18}{5})$
 E. $(3, 4)$ F. $(\frac{12}{5}, \frac{12}{5})$ G. $(\frac{12}{5}, 3)$

2. Y takes 0, 2, 4 with probabilities $1/4, 1/2, 1/4$. Define $W = 19 - 6Y$. Which pair gives $(E(W), \text{Var}(W))$?

- A. $(7, 72)$ B. $(7, 91)$ C. $(31, 72)$ D. $(12, 72)$
 E. $(7, -72)$ F. $(7, 12)$ G. $(7, 216)$

3. $\text{SD}(X) = 2$, $\text{SD}(Y) = 8$, and $\text{Corr}(X, Y) = -1/2$. Let $W = 5 + 2X$ and $V = 9 - 3Y$. Which pair gives $(\text{Cov}(W, V), \text{Corr}(W, V))$?

- A. $(-48, -\frac{1}{2})$ B. $(-8, -\frac{1}{2})$ C. $(8, \frac{1}{2})$ D. $(48, -\frac{1}{2})$
 E. $(93, \frac{1}{2})$ F. $(48, 3)$ G. $(48, \frac{1}{2})$

4. Three i.i.d. observations have mean μ and variance 36. Compare $A = (Y_1 + Y_2 + Y_3)/3$ and $B = Y_1/2 + Y_2/4 + Y_3/4$. Which pair gives $(\text{Var}(B), \text{Var}(B)/\text{Var}(A))$?

- A. $(\frac{27}{2}, \frac{8}{9})$ B. $(18, \frac{3}{2})$ C. $(12, 1)$ D. $(\frac{27}{2}, \frac{1}{8})$
 E. $(\frac{27}{2}, \frac{9}{8})$ F. $(36, 3)$ G. $(9, \frac{3}{4})$

5. Let Y_i be i.i.d. with mean μ and finite positive variance. Define $A_n = Y_1$, $B_n = \bar{Y}_n + 6/n$, and $C_n = \bar{Y}_n + 6$. Which statement correctly classifies all three rules for estimating μ ?

- A. A is unbiased and consistent; B is biased and inconsistent; C is biased and inconsistent.
 B. All three are inconsistent because finite-sample bias is nonzero.
 C. A is unbiased and inconsistent; B and C are biased and consistent.
 D. A is unbiased and inconsistent; B is biased and consistent; C is biased and inconsistent.
 E. All three are consistent because the nominal sample size grows.
 F. A is biased and inconsistent; B is unbiased and consistent; C is biased and inconsistent.
 G. A and B are unbiased and consistent; C is biased and inconsistent.

6. Four i.i.d. observations are 7, 9, 11, 13; the population shape is unspecified. Use $s_Y^2 = \sum_i (Y_i - \bar{Y})^2 / (n - 1)$ and $SE = s_Y / \sqrt{n}$. Which pair gives (s_Y^2, SE^2) ?

- A. $(\frac{5}{3}, \frac{20}{3})$ B. $(5, \frac{5}{4})$ C. $(\frac{20}{3}, \frac{5}{3})$ D. $(20, 5)$
 E. $(\frac{20}{3}, \frac{20}{3})$ F. $(\frac{20}{3}, \frac{5}{12})$ G. $(\frac{20}{3}, \frac{10}{3})$

7. An i.i.d. sample from a population with finite positive variance has $n = 144$, $\bar{y} = 26$ and sample SD $s_Y = 12$. Test $H_0 : \mu = 24$ against $H_1 : \mu \neq 24$ using the large-sample normal approximation. Which option gives the statistic, two-sided p-value, and decision at a prespecified 5% level? Use $\Phi(2) = 0.97725$.

- A. $t = -2$, $p \approx 0.0455$; reject. B. $t = 2$, $p \approx 0.0455$; do not reject.
 C. $t = 24$, $p \approx 0.0455$; reject. D. $t = 2$, $p \approx 0.9545$; do not reject.
 E. $t = 2$, $p \approx 0.02275$; reject. F. $t = 2$, $p \approx 0.0455$; reject.
 G. $t = 1/6$, $p \approx 0.0455$; reject.

8. OLS with an intercept uses $\bar{X} = 3$, $\bar{Y} = 25$, $S_{XX} = \sum_i (X_i - \bar{X})^2 = 10$, and $S_{XY} = \sum_i (X_i - \bar{X})(Y_i - \bar{Y}) = 60$. Which pair gives $(\hat{\beta}_0, \hat{\beta}_1)$?

- A. $(7, \frac{1}{6})$ B. $(6, 7)$ C. $(7, -6)$ D. $(43, 6)$
 E. $(7, 60)$ F. $(25, 6)$ G. $(7, 6)$

9. A simple OLS regression with an intercept uses $n = 18$ observations and reports $SSR = 576$ and $TSS = 2304$. Here SSR is the residual sum of squares. Which pair gives (R^2, SER) ?

- A. $(\frac{1}{3}, 6)$ B. $(\frac{3}{4}, 6\sqrt{16/17})$ C. $(\frac{3}{4}, 6)$ D. $(\frac{3}{4}, 6\sqrt{8/9})$
 E. $(\frac{3}{4}, 48)$ F. $(\frac{1}{4}, 6)$ G. $(\frac{3}{4}, 36)$

10. Maintain the causal model $Y = \alpha + 7X + 2Z + \varepsilon$, with $E(\varepsilon | X, Z) = 0$, $\text{Var}(X) = 4$ and $\text{Cov}(X, Z) = -2$. Pairs are sampled i.i.d. with finite fourth moments. Let $\hat{b}_s$ be the OLS slope on X alone, with an intercept. Which pair gives $(b_s, \text{plim } \hat{b}_s)$?

- A. $(7, 7)$ B. $(6, 6)$ C. $(3, 3)$ D. $(8, 8)$
 E. $(7, 6)$ F. $(6, 7)$ G. $(5, 5)$

Hard questions 11–20

11. The table lists joint probabilities, and $E(Y) = 3.50$.

	$Y = 0$	$Y = 7$	Total
$X = 0$	0.30	0.10	0.40
$X = 1$	a	b	0.60

Which triple gives $(a, b, \text{Pr}(X = 1 | Y = 7))$?

- A. $(\frac{1}{5}, \frac{2}{5}, \frac{5}{4})$ B. $(\frac{1}{5}, \frac{2}{5}, \frac{2}{3})$ C. $(\frac{1}{5}, \frac{2}{5}, \frac{1}{2})$ D. $(\frac{1}{5}, \frac{2}{5}, \frac{4}{5})$
 E. $(\frac{3}{10}, \frac{3}{10}, \frac{4}{5})$ F. $(\frac{1}{5}, \frac{2}{5}, \frac{2}{5})$ G. $(\frac{1}{5}, \frac{2}{5}, \frac{3}{5})$

12. For finite supports write $p_x = \text{Pr}(X = x) > 0$, $p_y = \text{Pr}(Y = y)$, and $p_{xy} = \text{Pr}(X = x, Y = y)$. A proof uses

$$E[E(Y | X)] = \sum_x \sum_y y \text{Pr}(Y = y | X = x) \boxed{1} = \sum_y y \sum_x \boxed{2} = E(Y).$$

Which choice supplies both blanks and states the required dependence assumption? Each replacement must be valid *term by term for every* (x, y) in an arbitrary joint distribution.

- A. (p_y, p_{xy}) ; independence is unnecessary.
 B. $(p_x, p_x p_y)$; independence is unnecessary.
 C. (p_x, p_y) ; independence is unnecessary.
 D. (p_x, p_{xy}) ; independence is required.
 E. $(1/p_x, p_{xy})$; independence is unnecessary.
 F. (p_x, p_{xy}) ; independence is unnecessary.
 G. $(1, \text{Pr}(Y = y | X = x))$; independence is unnecessary.

13. X takes $-7, -6, 6, 7$ with equal probabilities. Define $u = X^2 - \frac{85}{2}$. Which choice correctly gives $E(Xu)$, $E(X^2u)$, and whether $E(u | X) = 0$ holds?

- A. $(0, \frac{169}{4})$; zero conditional mean fails. B. $(0, \frac{169}{4})$; zero conditional mean holds.
 C. $(0, \frac{85}{2})$; zero conditional mean fails. D. $(0, -\frac{169}{4})$; zero conditional mean fails.
 E. $(0, \frac{169}{2})$; zero conditional mean fails. F. $(\frac{169}{4}, 0)$; zero conditional mean fails.
 G. $(0, 0)$; zero conditional mean holds.

14. Let Y_i be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. For $n > 4$, let $\bar{Y}_{-1}$ average observations $2, \dots, n$, and define

$$T_n = \frac{3}{4}Y_1 + \frac{1}{4}\bar{Y}_{-1}, \quad S_n = \frac{2}{\sqrt{n}}Y_1 + \left(1 - \frac{2}{\sqrt{n}}\right)\bar{Y}_{-1}.$$

Which pair gives $(\lim \text{Var}(T_n)/\sigma^2, \lim \text{Var}(S_n)/(\sigma^2/n))$ as $n \rightarrow \infty$?

- A. $(\frac{9}{16}, 1)$ B. $(\frac{9}{16}, 0)$ C. $(0, 1)$ D. $(0, 5)$
 E. $(\frac{9}{16}, 5)$ F. $(\frac{3}{4}, 5)$ G. $(\frac{9}{16}, 4)$

15. Independent Y_1, Y_2 are Bernoulli(0.70). Set $w_1 = 1$ if $Y_1 \geq Y_2$ and $w_1 = 0$ otherwise; $w_2 = 1 - w_1$. Write $T = w_1Y_1 + w_2Y_2$. A student claims

$$E(T) \stackrel{(1)}{=} \sum_{i=1}^2 E(w_i Y_i) \stackrel{(2)}{=} \sum_{i=1}^2 E(w_i) E(Y_i) \stackrel{(3)}{=} 0.70.$$

Which choice gives the actual $E(T)$ and the first invalid step?

- A. 0.91; step (2), because weights depend on outcomes.
 B. 0.49; step (2), because weights depend on outcomes.
 C. 0.91; step (2), because Y_1, Y_2 are dependent.
 D. 0.91; step (1), because expectation is not linear.
 E. 0.91; step (3), because weights sum to two.
 F. 0.70; no step is invalid.
 G. 0.30; step (2), because weights depend on outcomes.

16. Let Y_i be i.i.d. with mean μ and variance $0 < \sigma^2 < \infty$. Define $B_n = \bar{Y}_n - 5\sigma/\sqrt{n}$. Which pair gives its probability limit and the limiting distribution of $\sqrt{n}(B_n - \mu)/\sigma$? The second normal parameter is variance.

- A. $(\mu, \text{constant zero})$ B. $(\mu, N(-5, 1))$ C. $(\mu - 5\sigma, N(-5, 1))$ D. $(\mu, N(0, 1))$
 E. $(\mu, N(5, 1))$ F. $(\mu, \text{diverges to } -\infty)$ G. $(\mu, N(-5, 26))$

17. An i.i.d. sample from a normal population has $n = 324$, $\bar{y} = 16$ and known population SD $\sigma = 18$. Test $H_0 : \mu = 14$ against $H_1 : \mu \neq 14$. Each record is copied four times, creating 1296 rows from the same 324 independent draws. Which choice gives the *correct* z-statistic, two-sided p-value, and decision at 1% after copying? Use $\Phi(2) = 0.97725$ and $\Phi(4) = 0.9999683$.

- A. $z = 4, p \approx 0.0000634$; reject. B. $z = 2, p \approx 0.0455$; reject.
 C. $z = 2, p \approx 0.9545$; do not reject. D. $z = 4, p \approx 0.0455$; do not reject.
 E. $z = 2, p \approx 0.0455$; do not reject. F. $z = 2, p \approx 0.02275$; do not reject.
 G. $z = 1, p \approx 0.0455$; do not reject.

18. OLS with an intercept gives $\hat{Y} = 7 + 6X$ for

$$X = (0, 1, 2, 3), \quad Y = (13, a_*, b_*, 31).$$

Which triple gives (a_*, b_*, R^2) ?

- A. $(13, 7, \frac{5}{9})$ B. $(13, 19, 1)$ C. $(7, 13, 1)$ D. $(1, 19, \frac{5}{9})$
 E. $(7, 13, \frac{5}{4})$ F. $(7, 13, \frac{5}{9})$ G. $(7, 13, \frac{4}{9})$

19. For an arbitrary simple OLS sample *with an intercept*, assume $S_{XX} > 0$ and $TSS > 0$. Squaring $Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + \hat{u}_i$ gives

$$TSS = ESS + SSR + 2C, \quad C = \sum_i (\hat{Y}_i - \bar{Y}) \hat{u}_i.$$

Which expression for C, together with its justification for being zero, is valid for every such sample?

- A.** $\hat{\beta}_1 \sum_i (X_i - \bar{X})^2$; the intercept normal equation.
- B.** $\hat{\beta}_1 [\sum_i X_i u_i - \bar{X} \sum_i u_i]$; the sample normal equations for population errors.
- C.** $\sum_i \hat{u}_i^2$; the residuals have zero sample mean.
- D.** $\hat{\beta}_1 \sum_i X_i \hat{u}_i$; zero population conditional mean must be assumed.
- E.** $\hat{\beta}_1^2 S_{XX}$; the slope normal equation.
- F.** $\sum_i (Y_i - \bar{Y})^2$; an intercept makes total variation zero.
- G.** $\hat{\beta}_1 [\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$; both sample normal equations.

20. Maintain $Y = \alpha + \beta X + Z + \varepsilon$, with $E(\varepsilon | X, Z) = 0$, $SD(X) = SD(Z) = 2$, and $\text{Corr}(X, Z) = 1/2$. The population short-regression slope on X alone is $b_s = 5$. Under i.i.d. sampling and the lecture's finite-moment conditions,

$$\sqrt{n}(\hat{b}_s - b_s) \xrightarrow{d} N(0, 25).$$

Let $P_n = \Pr(|\hat{b}_s - \beta| \leq 1/4)$. Which triple gives $(\beta, P_{400}, \lim_{n \rightarrow \infty} P_n)$? Use the stated large-sample approximation, $\Phi(1) = 0.84134$ and $\Phi(3) = 0.99865$.

- | | | | |
|--|---|--|--|
| A. $(\frac{9}{2}, \approx 0.9987, 0)$ | B. $(5, \approx 0.6827, 1)$ | C. $(\frac{9}{2}, \approx 0.3173, 0)$ | D. $(\frac{9}{2}, \approx 0.1573, 0)$ |
| E. $(\frac{9}{2}, \approx 0.6827, 1)$ | F. $(\frac{11}{2}, \approx 0.1573, 0)$ | G. $(\frac{9}{2}, \approx 0.1573, 1)$ | |